CHARACTERIZATION OF W-CONTRACTIBLE AFFINE SCHEMES AS WEAKLY CONTRACTIBLE OBJECTS OF THE BIG PRO-ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the deduction of weak contractibility from the refinement property through finite Zariski covers and the induced product decompositions, and the deduction of w-contractibility by turning a faithfully flat weakly etale extension into a covering and producing the retraction from weak contractibility.

Editorial note. The label lemma-w-contractible-is-weakly-contractible names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-w-contractible-is-weakly-contractible –start 3026 –end 3084 –local-id 3582 returned RESULT=OK (L3582 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Equivalence of the ring theoretic condition with refinement of pro-etale coverings by finite Zariski coverings. Same chapter, the lemma identifying w-contractibility of an affine scheme with the property that every pro-etale covering is refined by a finite Zariski covering. Supplies the equivalence of parts (1) and (3).
- Construction of a faithfully flat weakly etale extension with a retraction. Same chapter, the proposition producing the extension A -> D of an affine w-local-like situation together with the retraction results used at the end. Supplies the object V = Spec(D) covering T used to produce the retraction sigma and hence the retractions of faithfully flat weakly etale extensions.
- Subcanonicity of the pro-etale topology and surjectivity of the representable maps. Same chapter, the lemma that the pro-etale topology is subcanonical, and Sites, the lemma turning a covering into a surjective map of sheaves after sheafification. Supply the surjectivity of h_V -> h_T, which combines with weak contractibility to produce a section of T over V.

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- The size of affine schemes and the bound on the big pro-etale site. Sets, the lemma computing the size of affine schemes, together with the definition of the big pro-etale site and the size estimate for the weakly contractible ring. Supplies the membership of V as an object of the big pro-etale site.

Source reference. Original reference 098G: sources/stacks-proetale/pinned-original-source.tex, source lines 2465–2465. Prerequisite pointer invoked at proof line(s) 3029; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 098B: sources/stacks-proetale/pinned-original-source.tex, source lines 2393–2393. Prerequisite pointer invoked at proof line(s) 3057; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 098Z: sources/stacks-proetale/pinned-original-source.tex, source lines 2966–2966. Prerequisite pointer invoked at proof line(s) 3073; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 098F: sources/stacks-proetale/pinned-original-source.tex, source lines 2998–2998. Prerequisite pointer invoked at proof line(s) 3044; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0983: sources/stacks-proetale/pinned-original-source.tex, source lines 2135–2135. Prerequisite pointer invoked at proof line(s) 3064; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0984: sources/stacks-proetale/pinned-original-source.tex, source lines 2171–2171. Prerequisite pointer invoked at proof line(s) 3067; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 000Q: sources/stacks-chow/tags.txt, source lines 35–35. Prerequisite pointer invoked at proof line(s) 3071; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 00WM: sources/stacks-chow/tags.txt, source lines 1188–1188. Prerequisite pointer invoked at proof line(s) 3052; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 090L: sources/stacks-chow/tags.txt, source lines 11414–11414. Prerequisite pointer invoked at proof line(s) 3035; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00WT: sources/stacks-chow/tags.txt, source lines 1195–1195. Prerequisite pointer invoked at proof line(s) 3075; it names a standard

fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $Sch_{pro\text{-}étale}$ be a big pro-étale site as in Definition 098G. Let $T = \mathrm{Spec}(A)$ be an affine object of $Sch_{pro\text{-}étale}$. The following are equivalent*

1. *A is w-contractible,*
2. *T is a weakly contractible (Sites, Definition 090L) object of $Sch_{pro\text{-}étale}$, and*
3. *every pro-étale covering of T can be refined by a Zariski covering of the form $T = \coprod_{i=1,\ldots,n} U_i$.*

Proof. We have seen the equivalence of (1) and (3) in Lemma 098F.

Assume (3) and let $\mathcal{F} \to \mathcal{G}$ be a surjection of sheaves on $Sch_{pro\text{-}étale}$. Let $s \in \mathcal{G}(T)$. To prove (2) we will show that s is in the image of $\mathcal{F}(T) \to \mathcal{G}(T)$. We can find a covering $\{T_i \to T\}$ of $Sch_{pro\text{-}étale}$ such that s lifts to a section of $\mathcal{F}$ over T_i (Sites, Definition 00WM). By (3) we may assume we have a finite covering $T = \coprod_{j=1,\ldots,m} U_j$ by open and closed subsets and we have $t_j \in \mathcal{F}(U_j)$ mapping to $s|_{U_j}$. Since Zariski coverings are coverings in $Sch_{pro\text{-}étale}$ (Lemma 098B) we conclude that $\mathcal{F}(T) = \prod \mathcal{F}(U_j)$. Thus $t = (t_1, \ldots, t_m) \in \mathcal{F}(T)$ is a section mapping to s.

Assume (2). Let $A \to D$ be as in Proposition 0983. Then $\{V \to T\}$ is a covering of $Sch_{pro\text{-}étale}$. (Note that $V = \mathrm{Spec}(D)$ is an object of $Sch_{pro\text{-}étale}$ by Remark 0984 combined with our choice of the function *Bound* in Definition 098G and the computation of the size of affine schemes in Sets, Lemma 000Q.) Since the topology on $Sch_{pro\text{-}étale}$ is subcanonical (Lemma 098Z) we see that $h_V \to h_T$ is a surjective map of sheaves (Sites, Lemma 00WT). Since T is assumed weakly contractible, we see that there is an element $f \in h_V(T) = \mathrm{Mor}(T, V)$ whose image in $h_T(T)$ is id_T. Thus $A \to D$ has a retraction $\sigma : D \to A$. Now if $A \to B$ is faithfully flat and weakly étale, then $D \to D \otimes_A B$ has the same properties, hence there is a retraction $D \otimes_A B \to D$ and combined with σ we get a retraction $B \to D \otimes_A B \to D \to A$ of $A \to B$. Thus A is w-contractible and (1) holds. □